

SOFT COMPUTING

ASSIGNMENT -8

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1.

a). Write a Python program to train a Back Propagation Neural Network (BPNN) for classifying handwritten digits.

```
import numpy as np
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical

(training_set_images, training_set_labels), (testing_set_images,
testing_set_labels) = mnist.load_data()

training_set_images = training_set_images.astype('float32') / 255.0
testing_set_images = testing_set_images.astype('float32') / 255.0

training_set_images = np.reshape(training_set_images,
(training_set_images.shape[0], 28 * 28))
testing_set_images = np.reshape(testing_set_images,
(testing_set_images.shape[0], 28 * 28))

training_set_labels = to_categorical(training_set_labels,
num_classes=10)
testing_set_labels = to_categorical(testing_set_labels,
num_classes=10)

model = models.Sequential()
model.add(layers.Input(shape=(28 * 28,)))
model.add(layers.Dense(512, activation='relu'))
```

```
model.add(layers.Dense(256, activation='relu'))
model.add(layers.Dense(10, activation='softmax'))

model.compile(optimizer='adam', loss='categorical_crossentropy',
metrics=['accuracy'])

epochs = 10
for epoch in range(epochs):
    print(f"Running Epoch {epoch + 1} out of {epochs}")
    history = model.fit(training_set_images, training_set_labels,
epochs=1, batch_size=32, validation_split=0.2, verbose=0)
    current_accuracy = history.history['accuracy'][0]
    print(f"Current Accuracy Here: {current_accuracy * 100:.2f}%")

testing_loss, testing_accuracy = model.evaluate(testing_set_images,
testing_set_labels)

print(f"The Accuracy of this BPNN Model Is : {testing_accuracy *
100:.2f}%")
```

```
Running Epoch 1 out of 10
Current Accuracy Here: 93.82%
Running Epoch 2 out of 10
Current Accuracy Here: 97.40%
Running Epoch 3 out of 10
Current Accuracy Here: 98.09%
Running Epoch 4 out of 10
Current Accuracy Here: 98.67%
Running Epoch 5 out of 10
Current Accuracy Here: 98.89%
Running Epoch 6 out of 10
Current Accuracy Here: 99.02%
Running Epoch 7 out of 10
Current Accuracy Here: 99.26%
Running Epoch 8 out of 10
Current Accuracy Here: 99.28%
Running Epoch 9 out of 10
Current Accuracy Here: 99.35%
Running Epoch 10 out of 10
Current Accuracy Here: 99.46%
313/313  0s 1ms/step - accuracy: 0.9725 - loss: 0.1294
The Accuracy of this BPNN Model Is : 97.57%
```

b). Write a Python program to train a Multi Layer Perceptron (MLP) for classifying handwritten digits. Compare the results of both neural networks.

```
import numpy as np
from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier

def load_data():
    mnist = fetch_openml('mnist_784')
    X, y = mnist['data'], mnist['target'].astype(int)
    return train_test_split(X, y, test_size=0.2, random_state=42)

def preprocess_data(X_train, X_test):
    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train)
    X_test_scaled = scaler.transform(X_test)
    return X_train_scaled, X_test_scaled

def train_mlp(X_train, y_train, epochs=10):
    mlp = MLPClassifier(hidden_layer_sizes=(128, 64),
        activation='relu', max_iter=100, warm_start=True, random_state=42)

    for epoch in range(epochs):
        mlp.fit(X_train, y_train)
        yield epoch + 1, mlp

def calculate_accuracy(model, X, y):
    predictions = model.predict(X)
    return np.mean(predictions == y) * 100

X_train, X_test, y_train, y_test = load_data()
X_train_scaled, X_test_scaled = preprocess_data(X_train, X_test)
```

```
epochs = 10
for epoch, model in train_mlp(X_train_scaled, y_train, epochs):
    current_accuracy = calculate_accuracy(model, X_train_scaled,
y_train)
    print(f"Running Epoch {epoch} out of {epochs}")
    print(f"Accuracy in this Epoch Is : {current_accuracy:.2f}%")

final_test_accuracy = calculate_accuracy(model, X_test_scaled,
y_test)

print(f"The Accuracy of this MLP Model Is :
{final_test_accuracy:.2f}%\n")
```

```
Running Epoch 1 out of 10
Accuracy in this Epoch Is : 100.00%
Running Epoch 2 out of 10
Accuracy in this Epoch Is : 99.83%
Running Epoch 3 out of 10
Accuracy in this Epoch Is : 99.82%
Running Epoch 4 out of 10
Accuracy in this Epoch Is : 99.81%
Running Epoch 5 out of 10
Accuracy in this Epoch Is : 99.78%
Running Epoch 6 out of 10
Accuracy in this Epoch Is : 99.86%
Running Epoch 7 out of 10
Accuracy in this Epoch Is : 99.92%
Running Epoch 8 out of 10
Accuracy in this Epoch Is : 99.83%
Running Epoch 9 out of 10
Accuracy in this Epoch Is : 99.90%
Running Epoch 10 out of 10
Accuracy in this Epoch Is : 99.90%
The Accuracy of this MLP Model Is : 97.33%
```